FIITJEE

ALL INDIA TEST SERIES

OPEN TEST – II

JEE (Advanced)-2021

PAPER – 1

TEST DATE: 21-03-2021

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. B

Sol. Given $|a_t| = 2|a_n|$

$-a_t = 2a_n$

$$-v\frac{dv}{ds} = 2\frac{v^2}{R}$$

$$-\int_{v_0}^{v}\frac{dv}{v} = \frac{2}{R}\int_{0}^{s} ds$$

$$\ell n\left(\frac{v}{v_0}\right) = -\frac{2s}{R}$$

$$v = v_0 e^{-\frac{2s}{R}}$$

When s = 10 m, $v = 20e^{-\frac{20}{10}} = 20e^{-2} = 20 \times 0.135 = 2.7$ m/s

V = 2.7 m/s

2. A

Sol. $v\cos 30° = \omega R\sqrt{3}\cos 60°$

$v = \omega R$ …(i)

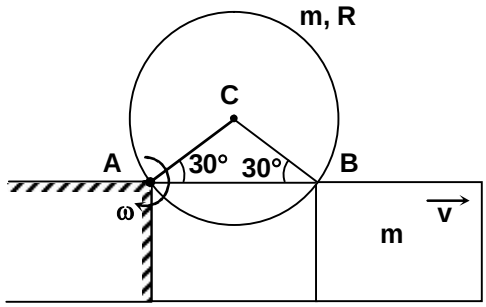
Now, using conservation of energy

$mgR(1-\sin 30°) = \frac{1}{2}\times 2mR^2\omega^2 + \frac{1}{2}mv^2$

$\frac{mgR}{2} = \frac{3}{2}mv^2$

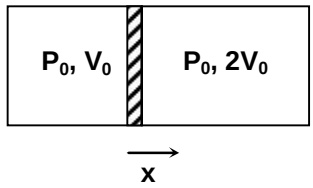
$\Rightarrow \quad gR = 3v^2$

$\Rightarrow \quad v = \sqrt{\frac{gR}{3}} = \sqrt{\frac{10\times 0.3}{3}} = 1\,m/s$



3. B

Sol. Since the ideal gas in each chamber undergoes an isothermal process



$\frac{dP}{dV} = -\left(\frac{P}{V}\right)$

$dP = -\left(\frac{P}{V}\right)dV$

When the piston is slightly displaced from equilibrium position by a distance 'x' the net restoring force acting on the piston is

$F = \left(\frac{P_0}{2V_0} + \frac{P_0}{V_0}\right)A^2x$

$F = \left(\frac{3P_0}{2V_0}\right)A^2x$

Now, $\frac{md^2x}{dt^2} = -F \Rightarrow \frac{md^2x}{dt^2} = -\left(\frac{3P_0}{2V_0}\right)A^2x \Rightarrow \frac{d^2x}{dt^2} = -\left(\frac{3P_0A^2}{2mv_0}\right)x$

$\therefore$ Time period, $T = 2\pi\sqrt{\frac{2mV_0}{3P_0A^2}}$

4. D

Sol. $I = \frac{Q\omega}{2\pi} = \frac{\sigma 2\pi R\ell\omega}{2\pi} = \sigma\omega R\ell$

The magnetic field produced inside the long hollow cylinder

$B = \frac{\mu_0 I}{\ell} = \mu_0\sigma\omega R$

$\int qEadt = ma^2\omega_0$

$qa\int\frac{a}{2}dB = ma^2\omega_0$

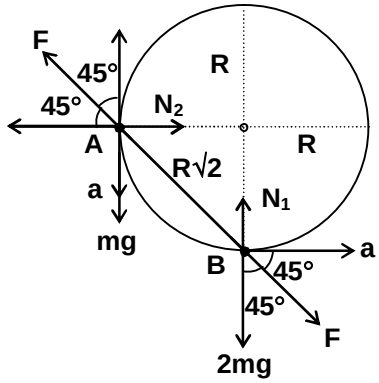
$\frac{qa^2}{2}B = ma^2\omega_0$

$\therefore \quad \omega_0 = \frac{qB}{2m} = \frac{q}{2m}(\mu_0\sigma\omega R) \quad \Rightarrow \quad \omega_0 = \frac{\mu_0 q\sigma\omega R}{2m}$

5. A, C, D

Sol. $m_A = m = 0.3$ kg

$m_B = 2m = 0.6$ kg



$N_1 = 2mg + \frac{F}{\sqrt{2}}$ …(i)

$\frac{F}{\sqrt{2}} = 2ma$ …(ii)

$N_2 = \frac{F}{\sqrt{2}}$ …(iii)

$mg - \frac{F}{\sqrt{2}} = ma$ …(iv)

From (ii) and (iv)

$mg = 3ma \Rightarrow a = g/3$

from (ii), $\frac{F}{\sqrt{2}} = 2ma = \frac{2mg}{3} = 2$

$\therefore F = 2\sqrt{2}$ N

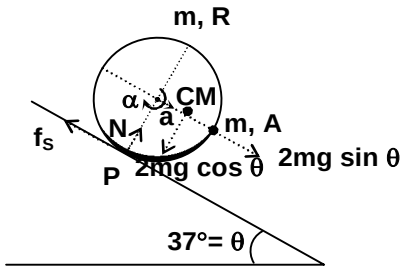
$\therefore N_2 = \frac{F}{\sqrt{2}} = 2N \Rightarrow N_2 = 2N$

From (i) $N_1 = 2mg + \frac{F}{\sqrt{2}} = 2mg + \frac{2mg}{3} = \frac{8mg}{3} = 8$ N

$\Rightarrow N_1 = 8N$

6. B, D

Sol. For pure rolling motion, $a = \alpha R$



$2mg\cos\theta \times \frac{R}{2} + 2mg\sin\theta \times R = 4mR^2\alpha$

$mg(\cos\theta + 2\sin\theta) = 4m\alpha R$

$g(\cos 37° + 2\sin 37°) = 4\alpha R$

$g\left(\frac{4}{5} + 2\times\frac{3}{5}\right) = 4\alpha R$

$2g = 4\alpha R \Rightarrow \alpha = \frac{g}{2R}$

$\therefore a = \alpha R = g/2$

Now, $2mg\sin\theta - f_S = 2ma$

$\frac{6mg}{5} - f_s = 2m\frac{g}{2} \Rightarrow f_s = \frac{6mg}{5} - mg = \frac{mg}{5} = 2N$

Also, $2mg\cos\theta - N = 2m\alpha\frac{R}{2}$

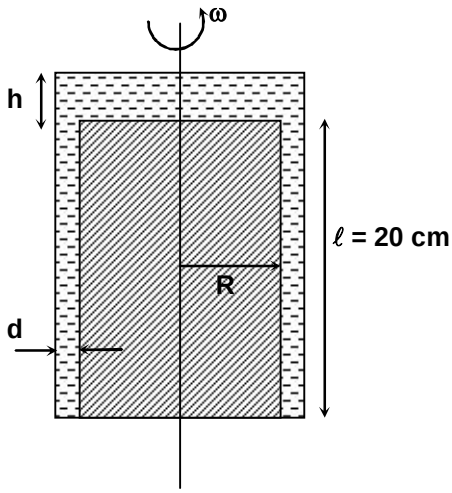
$\frac{8mg}{5} - N = \frac{mg}{2}$

$\Rightarrow N = \frac{8mg}{5} - \frac{mg}{2} = \frac{11}{10}mg$

N = 11N

7. A, D

Sol. The net torque of viscous force acting on the cylinder,



$$\tau = \int_0^R \eta 2\pi r dr \frac{\omega r}{h} r + \eta 2\pi R \ell \frac{\omega R}{d} R$$

$$= \frac{\pi\eta R^4 \omega}{2h} + \frac{2\pi\eta R^3 \ell\omega}{d} = \pi\eta R^3 \left(\frac{R}{2h} + \frac{2\ell}{d}\right)\omega$$

$$= \frac{22}{7} \times 7 \times 10^{-6} \left(\frac{10}{0.2} + \frac{40}{0.05}\right) 50$$

$\tau = 22 \times 10^{-6} \times 850 \times 50 = 0.935$ N-m

Power developed by the viscous force acting on the cylinder

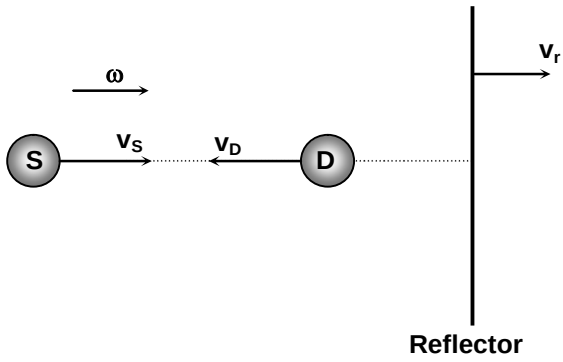
$P = \tau\omega = 0.935 \times 50 = 46.75$ Watt

P = 46.75 Watt

8. B, C

Sol. The frequency received by the reflector is

$$f_1 = \left(\frac{v + w - v_r}{v + w - v_s}\right) f$$



The frequency of the reflected wave received by the detector is

$$f_2 = \left(\frac{v - w - v_D}{v - w + v_r}\right) f_1$$

$$f_2 = \left(\frac{v - w - v_D}{v - w + v_r}\right)\left(\frac{v + w - v_r}{v + w - v_s}\right) f$$

$$= \left(\frac{330 - 10 - 10}{330 - 10 + 20}\right)\left(\frac{330 + 10 - 20}{330 + 10 - 30}\right) 527$$

$$f_2 = \left(\frac{310}{340}\right)\left(\frac{320}{310}\right) \times 527 = 496 \text{ Hz}$$

The wavelength of the reflected wave received by the detector is

$$\lambda_2 = \frac{v - w - v_D}{f_2} = \frac{330 - 10 - 10}{496} = \frac{5}{8} \text{ m}$$

9. A, C

Sol. $\frac{dy}{dx} = \tan i$

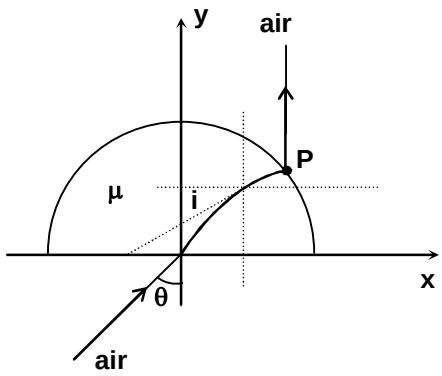
Now using snell's law

$1 \times \sin 90° = \mu \sin i$

$$\sin i = \frac{1}{\mu} = \left(\frac{a - x}{a}\right)$$

$$\sin i = \left(\frac{a - x}{a}\right)$$

$$\tan i = \frac{(a - x)}{\sqrt{a^2 - (a - x)^2}}$$



$\frac{dy}{dx} = \frac{(a-x)}{\sqrt{(2ax-x^2)}}$ from equation (i)

$\int_0^y dy = \int_0^x \frac{(a-x)}{\sqrt{2ax-x^2}} dx \Rightarrow y = \sqrt{2ax-x^2}$

Also, $y^2 + x^2 = \frac{a^2}{4}$ …(iii)

Solving (ii) and (iii) we get

$2ax = \frac{a^2}{4} \Rightarrow x = \frac{a}{8}$ and $y = \frac{a}{8}\sqrt{15}$

The x and y coordinates of point 'P' are $(\frac{a}{8}, \frac{a}{8}\sqrt{15})$

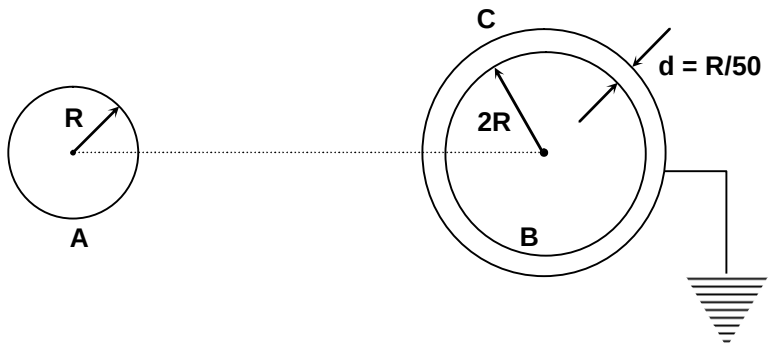
10. A, D

Sol. $C_1 = 4\pi\varepsilon_0 r_1 = 4\pi\varepsilon_0 R$

$C_2 = \frac{4\pi\varepsilon_0 r_2^2}{d} = \frac{4\pi\varepsilon_0 4R^2}{R/50} = 800\pi\varepsilon_0 R$

If charges appearing on the conducting spheres 'A' and 'B' are q_1 and q_2 respectively.

$\frac{q_1}{q_2} = \frac{C_1}{C_2} = \frac{4\pi\varepsilon_0 R}{800\pi\varepsilon_0 R}$



$\Rightarrow \frac{q_1}{q_2} = \frac{1}{200}$

$\Rightarrow q_2 = 200\, q_1$ …(i)

Now, $q_1 + q_2 = Q$

$q_1 + 200q_1 = Q$

$\Rightarrow q_1 = \frac{Q}{201}$

$q_2 = \frac{200Q}{201}$

11. A, C

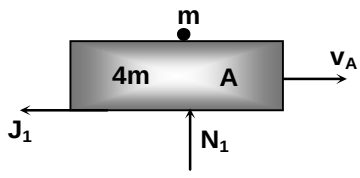
Sol. $\int N_1 dt = 8\ m$

Impulse due to friction on the block 'A'

$J_1 = \mu_1 \int N_1 dt = 0.4 \times 8m = 3.2\ m$ …(i)

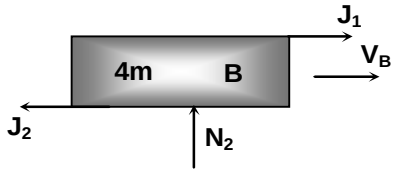
$J_1 = 6m - 5mv_A$

$3.2\ m = 6m - 5mv_A$



$\Rightarrow v_A = \frac{2.8}{5} = 0.56\,m/s$

$\int N_2 dt = 8m$



Impulse due to friction of horizontal surface on the block 'B'

$J_2 = \mu_2 \int N_2 dt = 0.2 \times 8$ m = 1.6 m ….(ii)

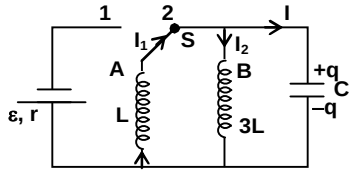
Now, $J_1 - J_2 = 4mv_B$

$\Rightarrow 4mv_B = 1.6$ m

$\Rightarrow v_B = 0.40$ m/s

12. B, C

Sol. $I_0 = \frac{\varepsilon}{r}$



$L\frac{dI_1}{dt} + 3L\frac{dI_2}{dt} = 0$

$\int_{I_0}^{I_1} dI_1 = -3\int_0^{I_2} dI_2$

$I_1 - I_0 = -3I_2$

$I_1 + 3I_2 = I_0$ …(i)

$I_1 - I_2 = I$ …(ii)

When the charge on the capacitor is maximum, I = 0

$I_1 = I_2$ …(iii)

From equation (1) and (iii)

$4I_2 = I_0 \Rightarrow I_2 = \frac{I_0}{4}$ and $I_1 = \frac{I_0}{4}$

Now, $\frac{1}{2}LI_0^2 - \frac{1}{2}(L+3L)\left(\frac{I_0}{4}\right)^2 = \frac{q_{max}^2}{2C}$

$LI_0^2 - \frac{LI_0^2}{4} = \frac{q_{max}^2}{C}$

$\Rightarrow \frac{3L}{4}I_0^2 = \frac{q_{max}^2}{C}$ $\Rightarrow q_{max} = \frac{I_0}{2}\sqrt{3LC} = \frac{\varepsilon}{2r}\sqrt{3LC}$

Now, when the current through the inductor 'B' is maximum, q = 0

$\frac{1}{2}LI_0^2 = \frac{1}{2}LI_1^2 + \frac{1}{2}3LI_2^2$

$I_0^2 = I_1^2 + 3I_2^2$

$\Rightarrow I_0^2 - I_1^2 = 3I_2^2$

$(I_0 + I_1)(3I_2) = 3I_2^2$

$\Rightarrow I_0 + I_1 = I_2$

$I_2 - I_1 = I_0$ …(iv)

Solving equation (i) and (iv) we get

$4I_2 = 2I_0 \Rightarrow I_2 = I_0/2 = \frac{\varepsilon}{2r}$

SECTION – C

13. 00020.25

Sol. Let the ball rebounds with a velocity 'v' after collision.

Using conservation of angular momentum about the axis AB.

$$mv_0\frac{2a}{3}=\frac{Ma^2}{3}\omega-\frac{mv2a}{3}$$

$$\frac{m2a}{3}(v_0+v)=\frac{Ma^2}{3}\omega$$

$v_0 + v = 2\,\omega a$ …(i)

$$v+\frac{2\omega a}{3}=v_0 \quad \text{…(ii)}$$

Solving (i) and (ii)

$v_0 + v = 3v_0 - 3v$

$4v = 2v_0 \Rightarrow v = v_0/2 = 3$ m/s

$$\Rightarrow\ \omega=\frac{v_0+v}{2a}=\frac{6+3}{2\times1}=4.5\ \text{rad/s}$$

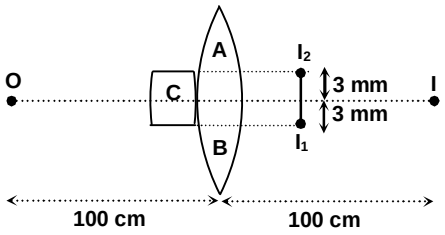
Horizontal component of the resultant force that will be exerted by the axis AB on the plate after the collision

$$F_x=M\omega^2\frac{a}{2}=2\times4.5\times4.5\times\frac{1}{2}=20.25\ \text{N}$$

$F_x = 20.25$ N

14. 00006.00

Sol. The image I of point object 'O' formed by the part 'C' is at a distance 100 cm right on the optic axis of the lens part 'C'.



Now,

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$$

$$\frac{1}{v}-\frac{1}{100}=\frac{1}{50}$$

$$v=+\frac{100}{3}\text{cm}$$

$$\therefore\ m=\frac{v}{u}=+\frac{1}{3}$$

$\therefore$ the distance between the two images formed by the parts A and B

d = 3 mm + 3mm = 6mm

15. 00013.50

Sol. $N_2 \sin\theta = (\rho - \rho_w)Va$ …(i)

$N_1 = N_2 \cos\theta + (\rho - \rho_w)Vg$

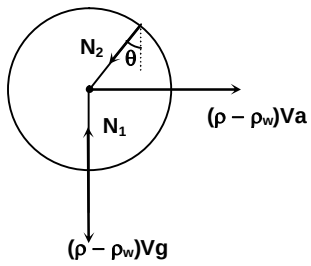
$N_1 = (\rho - \rho_w)$ Va cot $\theta + (\rho - \rho_w)Vg$

$N_1 = (\rho - \rho_w)V\,(g + a\cot\theta)$

$$N_1=(2500-1000)\times0.5\times10^{-3}\left(10+6\times\frac{4}{3}\right)$$

$= 1500 \times 0.5 \times 10^{-3} \times 18$

$N_1 = 13.50$ newton



16. 00009.00

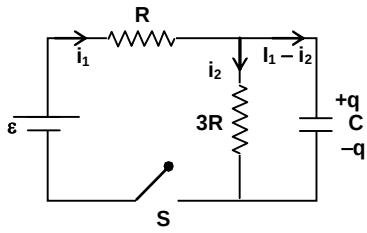
Sol. $i_1^2R = 16$ …(i)

$i_2^2 3R = 3$

$i_2^2R = 1$ …(ii)

From (i) and (ii)

$\frac{i_1^2R}{i_2^2R} = 16 \Rightarrow i_1 = 4i_2$ `



Energy stored in the capacitor at this instant

$U = \frac{q^2}{2C}$

$\frac{dU}{dt} = \frac{q}{C}\frac{dq}{dt} = (i_2 3R)(i_1 - i_2)$

$= (i_2 3R)3i_2 = 9i_2^2R = 9 \times 1 = 9$ watt

$\therefore \frac{dU}{dt} = 9$ watt

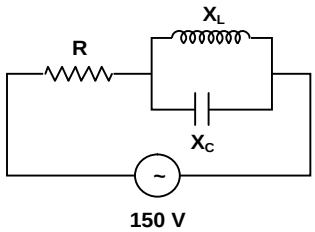
17. 00015.00

Sol. $Z_1 = X_L i = 8i$

$Z_2 = -X_C i = -4i$

$\frac{1}{Z'} = \frac{1}{Z_1} + \frac{1}{Z_2} = \frac{1}{8i} - \frac{1}{4i} = -\frac{i}{8} + \frac{i}{4}$

$\frac{1}{Z'} = \frac{i}{8} \Rightarrow Z' = \frac{8}{i} = -8i$



Net impedance of the circuit

$Z = R - 8i$

$Z = 6 - 8i$

$|Z| = \sqrt{(6)^2 + (8)^2} = 10\ \Omega$

rms current through the soruce, $I_{rms} = \frac{\varepsilon_{rms}}{|Z|} = \frac{150}{10} = 15$ A

rms current through $X_L = \frac{V_{rms}}{8}$

rms current through $X_C = \frac{V_{rms}}{4}$

Now, $\frac{V_{rms}}{4} - \frac{V_{rms}}{8} = 15$

$\frac{V_{rms}}{8} = 15 \Rightarrow V_{rms} = 120$ Volt

$\therefore$ rms current through $X_L = \frac{V_{rms}}{8} = 15$ A

18. 00001.10

Sol. ${}_1^2H + {}_1^2H \rightarrow {}_1^3H + {}_1^1P$

Using momentum conservation

$\vec{P}_1 + \vec{P}_2 = 0$

$\vec{P}_1 = -\vec{P}_2$

$P_1^2 = P_2^2$

$2m_1k_1 = 2m_2k_2$

$3k_1 = k_2$ …(i)

Also,

$k_1 + k_2 = 8.4 - 4$

$k_1 + k_2 = 4.4.$ …(ii)

solving (i) and (ii) we get

$4k_1 = 4.4.$

$k_1 = 1.1$ MeV and $k_2 = 3.3$ MeV

kinetic energy of the tritium, $k_1 = 1.1$ MeV

kinetic energy of the proton, $k_2 = 3.3$ MeV

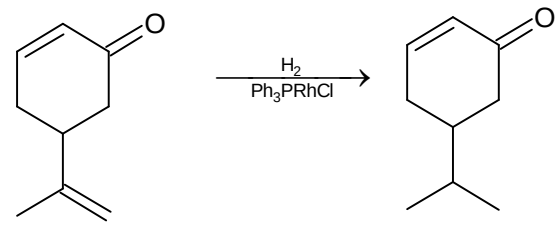
Chemistry

PART – II

SECTION – A

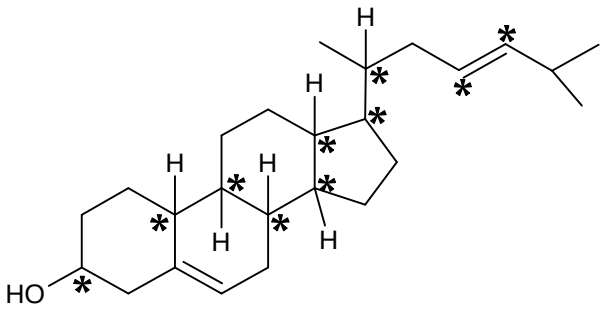
19. A
Sol. (A) is yellow while (B), (C) and (D) all are blue.

20. C
Sol.



So, A non-conjugated double bond is reduced more easily than a conjugated double bond by Wilkinson's catalyst.

21. D
Sol.



Where (*) represents a stereocentre.
So, total number of stereoisomers = 2^9.

22. B
Sol. A minimum of 6-H-atoms should be present in the sample, so that all the 10 spectral lines can be observed.

23. B, D
Sol. For an adiabatic expansion,

$$T \times V^{\gamma-1} = \text{Constant}(K)$$

Or $V \times T^{\frac{1}{\gamma-1}} = (K)^{\frac{1}{\gamma-1}} = \text{Constant}$

So, $\frac{1}{\gamma-1} = 3 \qquad \therefore \gamma = \frac{4}{3}$

So, gas must be non-linear polyatomic.

24. A, B, C, D
Sol. (i) $3A_2(g) \rightleftharpoons A_6(g)$

$$K_p = 1.6 = \frac{P_{A_6}}{(P_{A_2})^3} \Rightarrow P_{A_2} = \sqrt[3]{\frac{0.2}{1.6}} = 0.5 \text{ atm}$$

Also, pressure of A_2 used for the formation of A_6 = 0.6 atm.

$$A_2(g) + C(g) \rightleftharpoons A_2C(g)$$

(ii) at $t = 0$ 2P P 0

At eqbm: $(2P - p - 0.6)$ $(P - p)$ p

$$3A_2(g) \rightleftharpoons A_6(g)$$

Also, for at $t = 0$ 2P 0

At eqbm: $(2P - p - 0.6)$ 0.2

Since, P_{A_2} at equilibrium is 0.5 atm for simultaneous equilibria

Also, $P_{A_2} + P_C + P_{A_2C} + P_{A_6}$

$= (2P - p - 0.6) + (P - p) + p + 0.2 = 1.4$

P = 0.7 atm

Now, 2P – p – 0.6 = 0.5

1.4 – p – 0.6 = 0.5

p = 0.3 atm

So, $P_{A_2} = 0.5$ atm, $P_C = 0.7 - 0.3 = 0.4$ atm, $P_{A_2C} = 0.3$ atm.

Also, $A_2(g) + C(g) \rightleftharpoons A_2C(g)$

$$K_{p_2} = \frac{P_{A_2C}}{P_{A_2} \times P_C} = \frac{0.3}{0.5 \times 0.4} = 1.5 \text{ atm}^{-1}$$

25. A, B, C

Sol. Although 'P' exhibits four stereocentres, yet it exists only as one pair of enantiomers (due to increased symmetry)
In compound 'Q', the N-atom is at bridge head position, therefore, nitrogen inversion does not occur.
So, 'Q' is chiral.

26. A, B, D

Sol. Acetone and its next two homologues (2-butanone, 2-pentanone, 3-pentanone and isopropyl methyl ketone) reacts with hydroxyl amine to produce a total of eight oximes.

27. B, C

Sol. $$2S_2O_3^{2-} + \underset{(\text{neutral})}{Fe^{3+}} \longrightarrow \underset{\text{Violet complex}}{\left[Fe(S_2O_3)_2\right]^-} \xrightarrow{Fe^{3+}} \underbrace{2Fe^{2+} + S_4O_6^{2-}}_{(\text{Colourless})}$$

$$F_2 + \underset{(\text{excess})}{2KI} \longrightarrow 2KF + \underset{(\text{Violet})}{I_2} \xrightarrow{I^-} \underset{(\text{Brown})}{I_3^-}$$

$$Fe^{2+} + \left[Fe(CN)_6\right]^{3-} \longrightarrow \underset{\substack{(\text{Turnbull's blue}\\ \text{Permanent colour})}}{Fe\left[Fe(CN)_6\right]^-}$$

28. A, B, C, D

Sol. $H_2A + OH^- \longrightarrow HA^- + H_2O$

$HA^- + OH^- \longrightarrow A^{2-} + H_2O$

At first equivalent point, volume of base used is V_1 and at second equivalence point, it is V_2. So, $V_2 = 2V_1$.

When, $V = \frac{3V_1}{2}$, conc. of HA^- and A^{2-} will be equal and a buffer will be formed, so, $pH = pK_{a_2}$.

Also, when $V = \frac{V_1}{2}$, first buffer will be formed containing H_2A and A^- in equal concentration.

So, $pH = pK_{a_1}$.

29. A, C, D

Sol. $\underbrace{CrO_4^{2-} \text{ or } Cr_2O_7^{2-}}_{(X)} + H_2O_2 + H^+ \longrightarrow \underset{(Y)}{CrO_5} + H_2O$

$CrO_5 + H^+ \longrightarrow \underset{\left(\text{Green soln}\right)}{Cr^{3+}} + \underset{(M)}{O_2(g)} + H_2O$

30. B, C, D

Sol. Option B: Formula of solid would be $AB_2C_3D_8$.
Option C: Formula of solid would be $AB_4C_4D_8$.
Option A: Formula of solid would be $A_2B_6C_5D_{16}$.

SECTION – C

31. 00002.82

Sol. Mean free path $(\lambda) = \frac{1}{\sqrt{2}\pi d^2 N}$

Where, d = collision diameter of gas molecules.
N = Number of gas molecules per unit volume.

Average speed $= \sqrt{\frac{8RT}{\pi M}}$

32. 00000.40

Sol. $A \underset{k_b}{\overset{k_f}{\rightleftharpoons}} B(\text{fast})$

$B \xrightarrow[K_2]{} C(\text{slow})$

By writing steady state approximation for 'B'

$\frac{d[B]}{dt} = k_f[A] - k_b[B] - k_2[B] = 0$

$[B] = \frac{k_f[A]}{(k_b + k_2)}$

Now, rate $= k_2[B]$

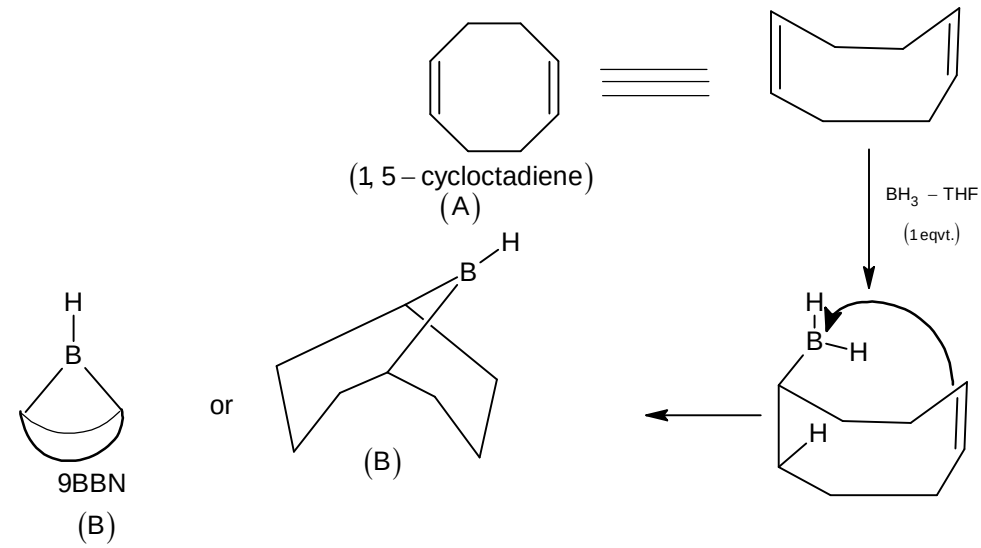
$r = \frac{k_2 \times k_f[A]}{(k_b + k_2)}$

So, $k = \frac{k_2 \times k_f}{(k_b + k_2)}$

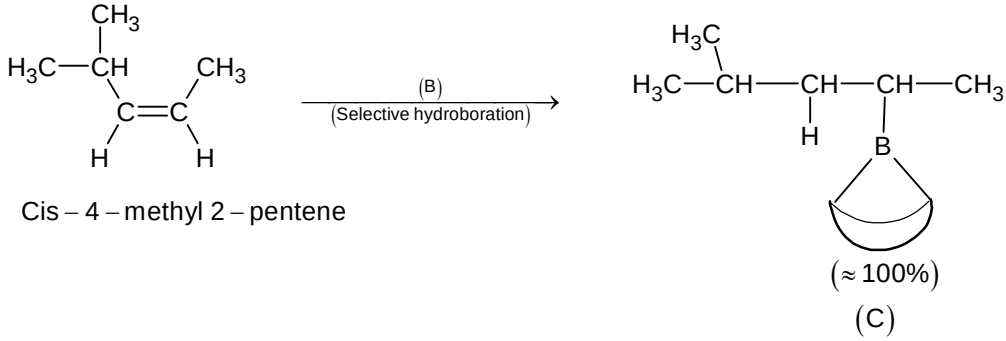
$k = \frac{2 \times 0.1}{0.4 + 0.1} = \frac{0.2}{0.5} = 0.40 \text{ sec}^{-1}$

33. 00002.50

Sol.



(9BBN = 9-Borabicyclo[3.3.1]nonane.)



Cis – 4 – methyl 2 – pentene

So, x = 2, y = 2, z = 0, m = 2

So, $\frac{xy+1}{(m+z)} = 2.50$

34. 00002.50

Sol. X = I_4O_9 $\underbrace{I(OIO_2)_3}_{\text{unionised form}} \longrightarrow \underbrace{I^{3+} + IO_3^-}_{\text{Ionised form}}$

So, m = 23, n = 8 and p = 3

35. 00003.00

Sol.



Number of resonating structure of $SO_4^{2-} = 6$

Number of plane of symmetry in $SO_4^{2-} = 6$

Number of resonating structure of $PO_4^{3-} = 4$

So, $\frac{6+6}{4} = 3$

36. 00010.40

Sol. $Fe_2S_3 + 5O_2 \longrightarrow 2FeSO_4 + SO_2$

So, equivalent weight of Fe_2S_3 = $= \frac{208}{20} = 10.40$ g

Mathematics

PART – III

SECTION – A

37. D

Sol. $|z^3 - z + 2|^2 = 16x^3 - 4x^2 - 16x + 8$, where $y^2 = 1 - x^2$, $x \in [-1, 1]$

So, max is $\sqrt{13}$

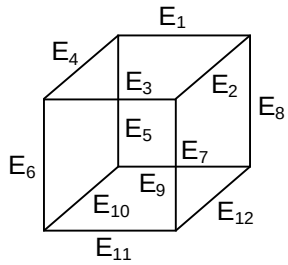
38. B

Sol. Let we can select two edges, from each vertex

$= {}^3C_2 \times 8 = 24$

From each parallel edges $= {}^4C_2 \times 3 = 18$

Total = 42



39. C

Sol. $1+\frac{p}{q} = \sum_{i=0}^{1000} \frac{{}^{1000}C_i}{{}^{2020}C_i} = \frac{1}{{}^{2020}C_{1000}} \sum_{i=0}^{1000} {}^{2020-i}C_{1020}$

$1+\frac{p}{q} = \frac{2021}{1021} \Rightarrow \frac{p}{q} = \frac{1000}{1021}$

40. C

Sol. $I = \int_0^{\pi/4} \frac{(\sec^2 x - 1)}{(\tan x - x)^2 + 3(\tan x - x) + 2} dx$. Let $\tan x - x = t \Rightarrow I = \int_0^{1-\frac{\pi}{4}} \frac{dt}{t^2 + 3t + 2} dx = \log_e\left(\frac{16 - 2\pi}{12 - \pi}\right)$

41. B, C

Sol. $\left[\vec{p}\ \vec{q}\ \vec{r}\right]^2 = \begin{vmatrix} \vec{p}\cdot\vec{p} & \vec{p}\cdot\vec{q} & \vec{p}\cdot\vec{r} \\ \vec{q}.\vec{p} & \vec{q}.\vec{q} & \vec{q}.\vec{r} \\ \vec{r}\cdot\vec{p} & \vec{r}.\vec{q} & \vec{r}.\vec{r} \end{vmatrix} = 9$, volume = $\frac{1}{2}$ unit^3. Base area = $\frac{1}{2}|\vec{p}\times\vec{q}|$, $|\vec{p}\times\vec{q}|^2 = |\vec{p}|^2|\vec{q}|^2 - (p\cdot q)^2$

42. B, C

Sol. Region traced by S is regular hexagon of side length 1 unit

43. A, B, D

Sol. Clearly $A_5 = 1$ with only possibility 01010 and $A_6 = 2$, with 011010 and 010110 only possible

Now, we have recursive function $A_n = A_{n-2} + A_{n-3}$ as 0 followed by either 10 or 110

44. A, D

Sol. If red balls is more than white balls, number ways $= {}^{13}C_3 \times {}^6C_3$

Number of ways such that white balls in exactly three boxes $= {}^4C_3 \times 3! \times {}^{20}C_3 \times {}^4C_1$

45. A, D

Sol. $\sum_{n=0}^{\infty} \frac{a_n b_n}{7^n} = \frac{1}{2} \text{Im}\left(\sum_{n=0}^{\infty}\left(\frac{3+4i}{7}\right)^n\right)$; $\sum_{n=0}^{\infty} \frac{a_n^2 - b_n^2}{7^n} = \text{Re}\left(\sum_{n=0}^{\infty}\left(\frac{3+4i}{7}\right)^n\right)$

46. A, B, C

Sol. $\Sigma\alpha^3 - 5\Sigma\alpha^2 + 8\Sigma\alpha = 39$; $S_3 - 5S_2 + 8S_1 = 39$, as $S_0 = 3$
So $S_3 = 5S_2 - 8S_1 + 13S_0$ and $S_4 = 5S_3 - 8S_2 + 13S_1$

47. A, B, C

Sol. The possible of roots can be either 2, 2, 2, 2ω, $2\omega^2$ or 2, 2ω, $2\omega^2$, 2ω, $2\omega^2$

48. A, D

Sol. Number of isosceles triangle are 30
Number of equilateral triangles whose at least two of the vertices belongs to set S = 66

SECTION – C

49. 00011.11

Sol. Let $N = \sum_{a=1}^{\infty}\sum_{b=1}^{\infty}\sum_{c=1}^{\infty}\frac{ab(3a+c)}{4^{a+b+c}(a+b)(b+c)(c+a)} \Rightarrow 6N = 3\sum_{a=1}^{\infty}\sum_{b=1}^{\infty}\sum_{c=1}^{\infty}\frac{1}{4^{a+b+c}}$

$N = \frac{1}{2}\left(\sum_{a=1}^{\infty}\frac{1}{4^a}\right)^3 = \frac{1}{54} \Rightarrow 600N = \frac{100}{9} = 11.11$

50. 00002.83

Sol. Let assume $a \geq b \geq c$; $(|a-b| + |b-c| + |c-a|)^2 \leq 4(a-c)^2 \leq 8 \Rightarrow |a-b| + |b-c| + |c-a| \leq 2\sqrt{2}$

51. 00036.00

Sol. $\left(z^{1^2} + z^{2^2} + z^{3^2} + ... + z^{12^2}\right)\left(\frac{1}{z^{1^2}} + \frac{1}{z^{2^2}} + \frac{1}{z^{3^2}} ..... + \frac{1}{z^{12^2}}\right)$ = 36

52. 00002.00

Sol. The area is 2 unit^2

53. 00072.00

Sol. Let f(x) – 12 = 0, has roots 1, 5, 6 with leading coefficients –2 as f(2) – 12 – 24 = –2(1)(–3) (–4)
So, $f(x) = -2(x-1)(x-5)(x-6) + 12 \Rightarrow |f(0)| = 72$
Similarly for f(x) – 12 = 0 with roots 2, 3 and 7

54. 00010.29

Sol. $\prod_{r=3}^{n}\frac{(r^3+3r)^2}{r^6-64} = \prod_{r=3}^{n}\frac{r}{r-2}\prod_{r=3}^{n}\frac{r}{r+2}\prod_{r=3}^{n}\frac{r^2+3}{(r+1)^2+3}\prod_{r=3}^{n}\frac{r^2+3}{(r-1)^2+3}$

So, $P(n) = \frac{n(n-1)}{2}\cdot\frac{12}{(n+1)(n+2)}\cdot\frac{12}{(n+1)^2+3}\cdot\frac{n^2+3}{7}$

$\lim_{n\to\infty} P(n) = \frac{72}{7}$